DIGITAL FORENSICS INVESTIGATION REPORT

Phishy Lab — Windows Disk Forensics

Case Name	Phishy Lab
Case Number	2026-006
Examiner	Tej Chaute
Date	03/04/2026
Platform	CyberDefenders
Evidence File	GiveAway.ad1 — SHA256: FE3707FE6163B2A23B26AC8D027A02F5472CB3F166C70E278C3CA096BEED10BB
Classification	Training

1. Case Summary

This investigation analyzes a phishing attack delivered via WhatsApp, where the victim was tricked into downloading a malicious document. The document contained VBA macros that executed a PowerShell command to download and run malware. The attack resulted in command-and-control communication and successful credential harvesting via a fake login page.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	GiveAway.ad1	FE3707FE6163B2A23B26AC8D027A02F5472CB3F166C70E278C3CA096BEED10BB	892 MB

3. Tools Used

Tool	Version	Purpose
Autopsy	4.22.1	Primary disk and artifact analysis
FTK Imager	4.7.3.81	Evidence mounting and preview
Registry Explorer	2.1.0	Registry hive analysis
DB Browser for SQLite	3.13.1.0	WhatsApp and browser database analysis
WhatsApp Viewer	1.15.0	Chat log reconstruction from msgstore.db
oledump.py	N/A	OLE document stream inspection and macro identification
olevba	N/A	VBA macro extraction and deobfuscation
CyberChef	10.22.1	Base64 decoding of obfuscated macro content
VirusTotal	2.5.8	Malware hash intelligence and C2 identification
PasswordFox	1.7.5.0	Firefox saved credential extraction

4. Key Findings

Each finding below is supported by at least one artifact. Path, timestamp, and source tool are documented for every item.

4.1 System Information

Field	Value
OS	Windows 7 Home
Computer name	WIN-NF3JQEU4G0T
User accounts	Semah
Timezone	IST (+5:30) — normalised to UTC during analysis

Field	Value
Last login	2021-03-19 22:53:11 UTC (SAM registry hive)

4.2 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	Registry	SYSTEM\ControlSet001\Control\ComputerName	--	Identifies the victim machine hostname (WIN-NF3JQEU4G0T), confirming the system under investigation
2	Executable File	Users\Semah\Downloads\WhatsApp.exe	2021-03-19 22:56:24	Indicates the presence of WhatsApp installer, supporting that the phishing attack was delivered via a messaging platform
3	WhatsApp Chat	Users/Semah/AppData/Roaming/WhatsApp/Databases/msgstore.db	2021-05-01 16:06:37	Shows attacker communication and delivery of phishing URL, establishing the initial access vector via social engineering
4	Malicious URL	http://apple.com/IPhone-Winners.doc	2021-03-19 20:43:49	Confirms delivery of a phishing document using a spoofed domain (apple.com), demonstrating a typosquatting attack
5	Document File	Users\Semah\Downloads\IPhone-Winners.doc	2021-05-01 17:56:52	Weaponised document downloaded — serves as first-stage malware delivery vehicle

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
6	Macro Stream	oledump stream 9 & 10 (IPhone-Winners.doc)	--	Document contains embedded VBA macros — confirms malicious nature and code execution capability
7	VBA AutoExec	Document_Open procedure (IPhone-Winners.doc)	--	Macro triggers automatically on document open via Document_Open — execution requires no user action beyond opening the file
8	PowerShell Execution	Decoded VBA macro payload (CyberChef)	--	Macro invokes obfuscated PowerShell command — confirms attacker-controlled code execution on victim system
9	Payload URL	http://apple.com/Iphone.exe	--	Second-stage payload download URL extracted from decoded macro — attacker-controlled infrastructure
10	File Write	C:\Temp\IPhone.exe	2021-05-01 20:04:24	Malicious executable written to disk — confirms successful second-stage payload delivery and staging
11	Malware Identification	VirusTotal (hash: 72C677BA...EBDBD1)	--	Payload identified as Metasploit-generated Meterpreter binary — confirms use of offensive framework for remote access

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
12	Network Artifact	Attacker C2 IP: 155.94.69.27 (identified via VirusTotal)	--	Outbound connection to attacker-controlled IP — confirms command-and-control channel established post-infection
13	Browser History	Users/Semah/AppData/Roaming/Mozilla/Firefox/Profiles/pyb51x2n.default-release/places.sqlite	2021-05-01 16:28:47	Victim visited fake Apple login page — confirms interaction with credential harvesting infrastructure
14	Phishing URL	http://apple.competitions.com/login.php	--	Fake Apple login page hosted on attacker domain — designed to harvest Apple account credentials
15	Stored Credentials	Users/Semah/AppData/Roaming/Mozilla/Firefox/Profiles/pyb51x2n.default-release/logins.json	2021-04-30 12:34:04	Victim credentials saved in Firefox after submission to phishing page — confirms successful credential compromise

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times in UTC.

Timestamp (UTC)	Artifact Source	Event Description
2021-03-19 20:43:49	WhatsApp Chat (msgstore.db)	Attacker delivered phishing URL http://apple.com/IPhone-Winners.doc to victim via WhatsApp
2021-03-19 22:56:24	File System (Downloads)	WhatsApp installer (WhatsApp.exe) present in Downloads — confirms communication platform

Timestamp (UTC)	Artifact Source	Event Description
2021-04-30 12:34:04	Browser Credentials (logins.json)	Victim credentials saved in Firefox — indicates prior visit to phishing login page
2021-05-01 16:06:37	WhatsApp Database (msgstore.db)	Chat database last modified — phishing conversation evidence preserved
2021-05-01 16:28:47	Browser History (places.sqlite)	Victim accessed phishing login page http://apple.competitions.com/login.php
2021-05-01 17:56:52	File System (Downloads)	Malicious document IPhone-Winners.doc downloaded to victim system
2021-05-01 ~	Document Execution	Victim opened IPhone-Winners.doc — VBA macro triggered automatically via Document_Open
2021-05-01 ~	Macro Execution	Obfuscated PowerShell command executed — initiated second-stage payload download from http://apple.com/Iphone.exe
2021-05-01 20:04:24	File System (C:\Temp)	Second-stage payload IPhone.exe written to disk — successful delivery confirmed
2021-05-01 ~	Network Activity	Meterpreter payload established outbound C2 connection to attacker IP 155.94.69.27

Note: The credential save timestamp (2021-04-30) predates the document download (2021-05-01). This may indicate the victim visited the phishing login page directly via the WhatsApp URL before downloading the document, or may reflect a timezone normalisation artefact. Both paths are consistent with the WhatsApp delivery vector confirmed in Artifact #3.

6. Conclusion

The investigation confirms that victim Semah on machine WIN-NF3JQEU4G0T was targeted through a multi-stage phishing attack delivered via WhatsApp (Artifact #3). The attacker used a typosquatted Apple domain (apple.com — capital 'I' in place of lowercase 'l') to deliver a malicious document (Artifact #4 and #5).

Upon opening the document, an embedded VBA macro (Artifact #6) triggered automatically via Document_Open (Artifact #7) and executed an obfuscated PowerShell command (Artifact #8) that downloaded a second-stage payload from the attacker's infrastructure (Artifact #9). The Meterpreter payload (Artifact #11) was written to C:\Temp\IPhone.exe (Artifact #10) and established a C2 connection to 155.94.69.27 (Artifact #12).

In parallel, the victim was redirected to a fake Apple login page at appIe.competitions.com (Artifact #14), where credentials were entered and subsequently saved by Firefox (Artifact #15), confirming successful credential compromise. The attack chain demonstrates a structured phishing campaign using multiple spoofed domains, social engineering via WhatsApp, and a Metasploit-based payload for persistent remote access.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path. Notes were partially recorded during the live investigation and completed after challenge submission.

Key methodological observations from this case:

- WhatsApp Viewer encountered a compatibility issue due to version mismatch with the msgstore.db format. The database was analysed directly using DB Browser for SQLite as a fallback, which successfully recovered the phishing conversation.
- oledump and olevba do not expose a --version parameter; versions are not independently verifiable through standard CLI invocation and are therefore listed as N/A.
- VBA macro content required Base64 decoding via CyberChef before the PowerShell payload URL could be extracted. The obfuscation layer (chr() concatenation + Base64) is a common technique to evade signature-based detection.
- The C2 IP address (155.94.69.27) was not directly recoverable from disk artifacts in this image. It was identified via VirusTotal and Hybrid Analysis using the IPhone.exe hash. In a live investigation, this would be corroborated through network PCAP or EDR telemetry.
- The typosquatted domain appIe.com (capital 'I' substituted for lowercase 'l') was used consistently across both the document delivery URL and the payload download URL, indicating a single attacker-controlled infrastructure.
- The credential save timestamp (2021-04-30) predates the document download (2021-05-01). This suggests the victim may have visited the phishing login page directly from the WhatsApp message link before opening the document, representing an independent credential harvesting path within the same campaign.

If this were a real-world investigation, recommended next steps would include: memory forensics to recover Meterpreter process artefacts and live C2 connection state; network PCAP analysis to confirm C2 beacon timing

and exfiltrated data; persistence mechanism analysis (Run keys, scheduled tasks); and full recovery of deleted files from unallocated space.